

Rafaela X. Solís (she/her)

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PROFESSIONAL SUMMARY

Gameplay and engine programmer pursuing dual B.S. degrees in Game Programming and Game Design with a Project Management minor. Works primarily in C++: building a custom physics engine, an OpenGL renderer with custom shaders, and Unreal Engine 5 mods in C++ and Lua, plus C# in Unity. Placed 2nd internationally in the CurseForge Subnautica 2 Modding Contest, with 6,000+ downloads across Subnautica 2 mods. Lead programmer at Pentad Games, building gameplay systems, CI pipelines, and team tooling. Design training means building systems that serve the player; production training means shipping them on schedule.

EDUCATION

Champlain College

*Bachelor of Science in Game Programming; Bachelor of Science in Game Design
Project Management Minor. GPA: 3.90. Dean's List, all semesters.*

Burlington, VT
Expected May 2028

Relevant Coursework: Game Architecture, Game Physics, Artificial Intelligence for Games, Introduction to Modern Graphics Programming, Data Structures and Algorithms, Advanced Programming, Matrices, Vectors, and 3D Math, Discrete Mathematics

EXPERIENCE

Lead Developer (Lead Programmer, Lead Designer, Product Owner)

October 2024 – Present

Pentad Games (formerly Vellum Games)

- Lead programming for a 5–7 person team (up to 10) on Open Shift, a co-op cozy-horror storekeeping game entering beta, and Scriptorium; served as engine implementer on Mega Market Mayhem
- Designed and implemented 50+ gameplay features in Unity 6 (C#), shipping weekly playable builds within Agile/Scrum
- Built a GitHub Actions CI pipeline for nightly Unity builds and a repo integration to browse files and download builds or source
- Trained multiple CNN iterations in Python (TensorFlow, Keras) to recognize player-drawn runes, retraining on playtester samples
- Built and self-host the studio's AFFiNE workspace: kanban tracking story points, assignees, and completion, plus email reminders
- Developed local AI meeting transcription and note-taking, an art hub with automated page generation and critique workflows, and a Discord bot integrated with Taiga for weekly sprint breakdowns

Project Manager

October 2026 – Present

Leahy Center for Digital Forensics & Cybersecurity, Champlain College

Burlington, VT

- Restructure documentation, workflows, and meetings across the organization's teams while keeping each team's familiar setup

Producer

August 2024 – Present

Champlain College Academic Game Projects

- Produced 5+ multidisciplinary teams of 3–6 on weekly sprints, shipping every project on time across one-month to multi-year runs
- Author technical and production documentation that aligns programming with art, design, and audio

PROJECTS

C++ Engine and Graphics Programming

August 2024 – Present

- Building Cyclone, a custom C++ physics engine (Raylib, Dear ImGui) with particle simulation and frame-rate-independent damping
- Built a custom OpenGL renderer with hand-written GLSL shaders and Phong lighting, plus a full game written entirely in C++

Subnautica 2 Modding Contest, 2nd Place Internationally (CurseForge)

June 2026

- Built Easy Third Person, a third-person camera mod in C++ and Lua for Unreal Engine 5; won 2nd place internationally and \$650
- Reverse-engineered UE4SS to implement custom systems; added camera collision after playtesters saw it clip through walls
- Released Easy Third Person and The Gold Mod (6,000+ combined downloads); featured on CurseForge's main banner for weeks

Indie Game Development <https://solunabeeboo.itch.io/>

March 2020 – Present

- Designed and programmed 12+ original games solo, owning systems, art, build, and release
- Built and published 3 Minecraft modpacks on CurseForge with 560+ combined downloads

SKILLS

Languages: C++, Lua, C#, Python, Java, HTML/CSS

Engines and Frameworks: Unreal Engine 5 (Lyra Framework), UEFN, UE4SS, Blueprints, Unity 6, OpenGL, GLSL Shaders, Raylib, Dear ImGui, TensorFlow, Keras

Tools and Pipeline: Git, GitHub Actions (CI), Subversion, CMake, Self-Hosted Services, Jira, Confluence, Taiga, AFFiNE

Game Development: Engine Programming, Physics Programming, Graphics Programming, Gameplay Programming, Systems Design, Technical Design, Tools Development, Rapid Prototyping, Playtesting and Data Analysis, Agile/Scrum

Art: Blender, Aseprite, Photoshop, Pixel Art, 3D Modeling